# A Theoretical Study of the Finite Element Method for the Two-Dimensional Steady-State Heat Equation

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**Abstract**

This report presents a detailed theoretical study of the Finite Element Method (FEM) applied to the steady-state heat conduction equation in two dimensions. The focus of this work is on developing a clear understanding of the physical principles, mathematical formulation, and variational foundations that lead to the finite element discretization of the governing partial differential equation. Emphasis is placed on conceptual clarity and rigorous derivation rather than numerical implementation, with the goal of building a strong foundation for future computational and research-oriented applications.

# 1 Aim of the Project

The aim of this personal project is to develop a complete and rigorous theoretical understanding of the Finite Element Method (FEM) through its application to the steady-state heat conduction equation in two dimensions. Rather than treating FEM as a purely numerical or algorithmic tool, the objective of this work is to study FEM as a physically motivated and mathematically consistent framework that arises naturally from fundamental principles of physics and calculus.

This project seeks to trace the full logical path from physical laws governing heat transfer to the final matrix equation solved in a finite element simulation. Particular emphasis is placed on understanding the origin of the governing partial differential equation, the limitations of classical analytical solution methods, and the motivation behind reformulating the problem into a weak or variational form. By carefully deriving each step and interpreting its physical meaning, this project aims to eliminate the conceptual gap that often exists between mathematical formalism and physical intuition in numerical methods.

In addition, this work serves as a foundation for future computational research involving more complex systems, such as time-dependent heat transfer, coupled multi-physics problems, and partial differential equations arising in astrophysics, nuclear physics, and materials science. The focus of the present report is therefore on conceptual clarity, mathematical rigor, and physical interpretation.

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# 2 Introduction

Heat conduction is one of the most fundamental transport phenomena in physics and engineering. It governs the transfer of thermal energy in solids and plays a crucial role in applications ranging from electronic device design and structural engineering to planetary science and nuclear reactor analysis. In many realistic physical systems, heat transfer occurs in domains with irregular geometries, non-uniform material properties, and complex boundary conditions, which significantly complicates analytical treatment.

The mathematical description of heat conduction is given by partial differential equations derived from conservation laws and constitutive relations. While exact analytical solutions exist for a limited class of idealized problems, such as infinite or highly symmetric domains, most practical problems cannot be solved in closed form. This limitation motivates the use of numerical methods, which approximate the solution by discretizing the problem in space and, when necessary, in time.

Among various numerical approaches, the Finite Element Method has emerged as one of the most powerful and versatile techniques for solving partial differential equa-

tions in complex geometries. Unlike point-based methods such as finite differences, FEM is built upon an integral formulation of the governing equations and is closely tied to variational principles and energy minimization. This makes FEM particularly well suited for problems involving conservation laws and boundary-driven physics.

The purpose of this project is to study the finite element method from first principles by applying it to the steady-state heat equation in two dimensions. The emphasis is placed on understanding how the governing equation is derived from physical laws, why the classical differential (strong) form is unsuitable for direct numerical approximation, and how the weak form naturally leads to the finite element discretization. The work presented here focuses entirely on the theoretical formulation, which forms the essential foundation for any correct and reliable numerical implementation.

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# 3 Physical and Mathematical Formulation of the Heat Equation

## 3.1 Temperature as a Field Variable

In heat conduction problems, temperature is not a single scalar quantity but a spatially varying field. In a two-dimensional domain $\Omega \subset \mathbb{R}^2$, the temperature is described by a function

$$T = T(x, y),$$

which assigns a temperature value to every point within the domain. The objective of a heat conduction analysis is to determine this temperature distribution under specified physical conditions.

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## 3.2 Fourier's Law of Heat Conduction

Heat flows in response to temperature gradients. This physical behavior is described by Fourier's law of heat conduction, which states that the heat flux vector $\mathbf{q}$, representing the rate of heat flow per unit area, is proportional to the negative gradient of temperature:

$$\mathbf{q} = -k\nabla T,$$

where $k$ is the thermal conductivity of the material. The negative sign indicates that heat flows from regions of higher temperature to regions of lower temperature.

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## 3.3 Energy Conservation and Governing Equation

The governing equation for heat conduction is obtained by applying the principle of conservation of energy to a differential control volume within the domain. Under steady-state conditions, the temperature does not vary with time, and the net heat entering a control volume must balance the heat leaving it along with any internally generated heat. This balance is expressed mathematically as:

$$\nabla \cdot \mathbf{q} + Q = 0,$$

where $Q$ denotes the internal heat generation per unit volume.

Substituting Fourier's law into this expression yields:

$$\nabla \cdot (-k\nabla T) + Q = 0.$$

Assuming constant thermal conductivity, this equation simplifies to:

$$-k\nabla^2 T = Q.$$

Dividing by $k$, the steady-state heat equation takes the form:

$$-\nabla^2 T = f(x, y),$$

where $f(x, y) = Q/k$ represents the normalized heat source term.

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## 3.4 Strong Form of the Heat Equation

The equation

$$-\nabla^2 T = f(x, y)$$

is referred to as the *strong form* of the governing equation. In this formulation, the equation must be satisfied pointwise at every location within the domain $\Omega$. This requires the temperature field to possess continuous second-order spatial derivatives, a condition that is difficult to enforce numerically in complex geometries.

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## 3.5 Motivation for the Weak Formulation

To overcome the limitations of the strong form, the governing equation is reformulated into a weak or variational form. This is achieved by multiplying the strong form by an

admissible test function $v$ and integrating over the domain:

$$\int_\Omega (-\nabla^2 T) v \, d\Omega = \int_\Omega f v \, d\Omega.$$

Applying integration by parts transfers the second-order derivatives from the temperature field to the test function, resulting in:

$$\int_\Omega \nabla T \cdot \nabla v \, d\Omega = \int_\Omega f v \, d\Omega,$$

for the case of prescribed temperature (Dirichlet) boundary conditions. This weak formulation reduces the differentiability requirements on the solution and provides the mathematical foundation for the finite element method.

# 4 Variational (Weak) Formulation of the Heat Equation

The finite element method is fundamentally based on a variational or weak formulation of the governing differential equation. This formulation plays a central role in enabling numerical approximation while preserving the underlying physical principles of the problem. In this section, the weak form of the steady-state heat equation is derived and its significance is discussed in detail.

## 4.1 From Strong Form to Weak Form

The strong form of the steady-state heat equation is given by

$$-\nabla^2 T = f \quad \text{in } \Omega,$$

where $T(x, y)$ denotes the temperature field and $f(x, y)$ represents the heat source term. In the strong formulation, this equation is required to hold pointwise at every location within the domain $\Omega$. Such a requirement imposes strict smoothness conditions on the solution, namely the existence of continuous second-order derivatives, which are difficult to satisfy in numerical approximations over complex geometries.

To relax these requirements, the governing equation is reformulated into a weak form. This is achieved by multiplying the strong form by an admissible test function $v$, which vanishes on portions of the boundary where temperature is prescribed, and integrating over the domain:

$$\int_\Omega (-\nabla^2 T) \, v \, d\Omega = \int_\Omega f \, v \, d\Omega.$$

## 4.2 Application of Integration by Parts

The integral on the left-hand side contains second-order derivatives of the temperature field. To reduce the order of differentiation, integration by parts is applied. Using Green's identity, the following transformation is obtained:

$$\int_\Omega (-\nabla^2 T)\, v \, d\Omega = \int_\Omega \nabla T \cdot \nabla v \, d\Omega - \int_{\partial\Omega} v \, \frac{\partial T}{\partial n} \, dS,$$

where $\partial\Omega$ denotes the boundary of the domain and $\frac{\partial T}{\partial n}$ is the normal derivative of the temperature.

For problems with prescribed temperature (Dirichlet boundary conditions), the test function $v$ is chosen to vanish on the boundary, causing the boundary integral to disappear. Under this assumption, the governing equation reduces to:

$$\int_\Omega \nabla T \cdot \nabla v \, d\Omega = \int_\Omega f \, v \, d\Omega.$$

## 4.3 Final Weak Formulation

The above expression constitutes the weak or variational form of the steady-state heat equation. In this formulation, the governing equation is no longer enforced at individual points but rather in an integral sense over the domain. This significantly reduces the smoothness requirements on the temperature field, allowing it to be approximated using piecewise-defined functions.

From a physical perspective, the weak formulation represents a global balance of thermal energy, ensuring that the integral contribution of heat flow and internal sources is consistent throughout the domain. This variational structure forms the mathematical foundation of the finite element method and enables the systematic construction of discrete approximation spaces in subsequent sections.

The weak formulation derived here serves as the starting point for finite element discretization, where both the temperature field and the test functions are approximated using suitable basis functions defined over finite elements.

# 5 Finite Element Discretization

The finite element method is based on the idea of approximating a continuous physical problem by dividing the domain into smaller, simpler parts. Instead of solving the governing equation over the entire domain at once, the problem is solved approximately over many small subdomains called elements. This approach makes it possible to handle complex geometries and boundary conditions in a systematic and stable manner.

## 5.1 Discretization of the Computational Domain

Let $\Omega \subset \mathbb{R}^2$ denote the two-dimensional physical domain in which heat conduction takes place. In the finite element method, this domain is divided into a finite number of non-overlapping triangular elements such that

$$\Omega = \bigcup_{e=1}^{N_e} \Omega_e,$$

where $\Omega_e$ represents the region occupied by the $e$-th element and $N_e$ is the total number of elements in the mesh.

The corners of each triangular element are called nodes. These nodes form the primary points at which the temperature is computed. Neighboring elements share nodes along their edges, which ensures continuity of the temperature field across the entire domain. Triangular elements are chosen in this study because they can easily conform to irregular geometries and allow a simple and robust mathematical formulation in two dimensions.

## 5.2 Approximation of the Temperature Field

In the finite element framework, the exact temperature field $T(x, y)$ is replaced by an approximate temperature field $T_h(x, y)$. This approximation is constructed using a linear combination of predefined basis functions, known as shape functions:

$$T_h(x, y) = \sum_{i=1}^{N} T_i \, \phi_i(x, y),$$

where $N$ is the total number of nodes in the mesh, $T_i$ denotes the unknown temperature at node $i$, and $\phi_i(x, y)$ is the shape function associated with that node.

Each shape function represents the influence of a particular node on the temperature at any point within the domain. The shape functions are constructed such that they satisfy the interpolation property

$$\phi_i(\mathbf{x}_j) = \delta_{ij},$$

which ensures that the approximate solution exactly matches the nodal temperature values at the nodes.

## 5.3 Choice of Linear Triangular Shape Functions

In this study, linear shape functions defined over triangular elements are used. Within a single triangular element, the temperature is assumed to vary linearly with spatial coordinates. This assumption leads to constant temperature gradients inside each element, which significantly simplifies both the theoretical formulation and the numerical

implementation.

Although higher-order shape functions can provide greater accuracy, linear triangular elements offer an excellent balance between simplicity, stability, and physical interpretability, making them an ideal choice for a foundational study of the finite element method.

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# 6 Element-Level Formulation

Having defined the finite element approximation, the next step is to derive the governing equations at the level of an individual element. These element-level equations describe how heat is conducted within a single triangular element and form the fundamental building blocks of the global finite element system.

## 6.1 Geometry of a Linear Triangular Element

Consider a triangular element $\Omega_e$ with three vertices located at coordinates

$$(x_1, y_1), \quad (x_2, y_2), \quad (x_3, y_3).$$

The area of this triangle is given by

$$A_e = \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}.$$

The geometric properties of the element, such as its area and shape, play a direct role in determining how heat flows within the element.

## 6.2 Linear Shape Functions Within an Element

For a linear triangular element, the shape functions are assumed to be linear functions of the spatial coordinates:

$$\phi_i(x, y) = a_i + b_i x + c_i y, \quad i = 1, 2, 3.$$

The coefficients $a_i, b_i, c_i$ are uniquely determined by enforcing the condition that each shape function equals unity at its own node and zero at the other two nodes. This construction ensures that the temperature field inside the element is completely determined by the nodal temperature values.

## 6.3 Gradient of the Shape Functions

A key property of linear triangular elements is that the gradient of each shape function is constant within the element. The gradient is given by

$$\nabla \phi_i = \begin{bmatrix} \frac{\partial \phi_i}{\partial x} \\ \frac{\partial \phi_i}{\partial y} \end{bmatrix} = \frac{1}{2A_e} \begin{bmatrix} b_i \\ c_i \end{bmatrix}.$$

This constant-gradient property greatly simplifies the evaluation of integrals appearing in the weak formulation.

## 6.4 Derivation of the Element Stiffness Matrix

Substituting the finite element approximation into the weak formulation and restricting the integration to a single element leads to the element stiffness matrix, defined as

$$K_{ij}^{(e)} = \int_{\Omega_e} \nabla \phi_i \cdot \nabla \phi_j \, dA.$$

Since the gradients of the shape functions are constant within the element, the integrand is also constant, and the integral can be evaluated analytically. This results in

$$K_{ij}^{(e)} = \frac{1}{4A_e}(b_i b_j + c_i c_j).$$

The element stiffness matrix is symmetric and represents the resistance to heat flow within the element.

## 6.5 Element Load Vector

The contribution of internal heat generation within an element is captured by the element load vector, defined as

$$F_i^{(e)} = \int_{\Omega_e} f \, \phi_i \, dA.$$

For a constant heat source $f$, this integral simplifies to

$$F_i^{(e)} = \frac{f A_e}{3}.$$

This expression indicates that the heat generated within an element is distributed equally among its three nodes.

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# 7 Assembly and Boundary Conditions

The element-level equations derived in the previous section must be combined to obtain a system of equations that describes heat conduction over the entire domain. This process is known as assembly and is followed by the application of appropriate boundary conditions.

## 7.1 Assembly of the Global System

Each triangular element contributes a local stiffness matrix and a local load vector. These local quantities are assembled into a global stiffness matrix $K$ and a global load vector $\mathbf{F}$ using the connectivity information of the mesh. This mapping ensures that contributions from elements sharing a common node are added together.

Mathematically, the global system is obtained by summing the contributions of all elements:

$$K = \sum_{e=1}^{N_e} K^{(e)}, \quad \mathbf{F} = \sum_{e=1}^{N_e} \mathbf{F}^{(e)}.$$

This assembly process enforces continuity of the temperature field across element boundaries and ensures global conservation of thermal energy.

## 7.2 Imposition of Boundary Conditions

Boundary conditions are essential for obtaining a unique and physically meaningful solution. In heat conduction problems, two types of boundary conditions are commonly encountered. Dirichlet boundary conditions prescribe the temperature on a portion of the boundary, while Neumann boundary conditions prescribe the heat flux.

Prescribed temperature boundary conditions are enforced directly by fixing the corresponding nodal temperature values in the global system of equations. Prescribed heat flux boundary conditions contribute additional terms to the global load vector through boundary integrals.

After assembling the system and applying the boundary conditions, the final system of equations takes the form

$$K\mathbf{T} = \mathbf{F},$$

where $\mathbf{T}$ is the vector of unknown nodal temperatures. Solving this linear system yields the finite element approximation of the temperature distribution over the domain.

# 8 Numerical Results and Analysis

## 8.1 Constant Heat Source $f = 1$

### 8.1.1 Temperature Distribution

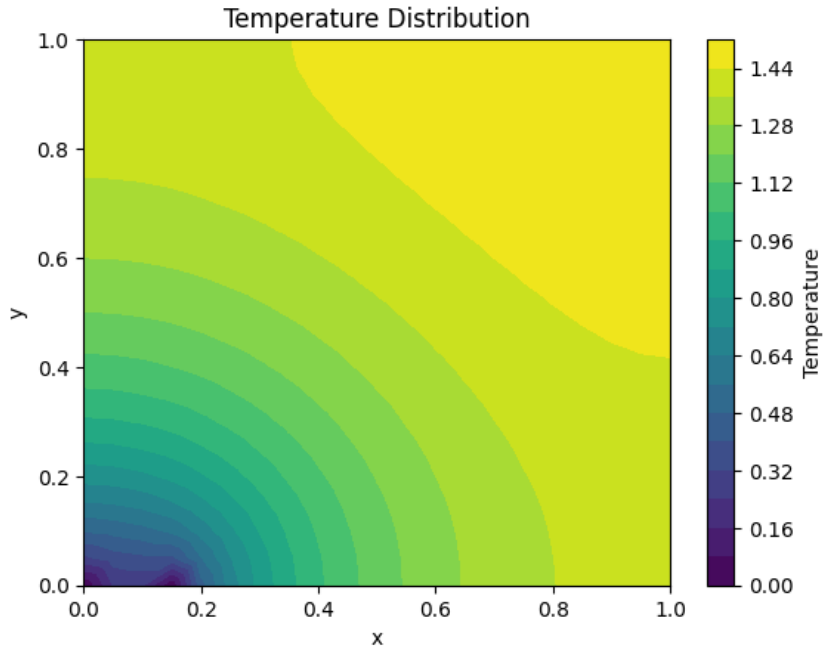


Figure 1: Temperature distribution for constant heat source $f = 1$.

The temperature field for the constant heat source exhibits a smooth and continuous variation across the domain. Since heat generation is uniform, the resulting temperature profile reflects a balance between internal heat production and diffusion toward the boundaries.

There are no localized peaks or sharp variations, indicating that the system is governed by global thermal behavior. This result is consistent with the expected solution of the steady-state heat equation under uniform forcing.

### 8.1.2 Gradient Magnitude

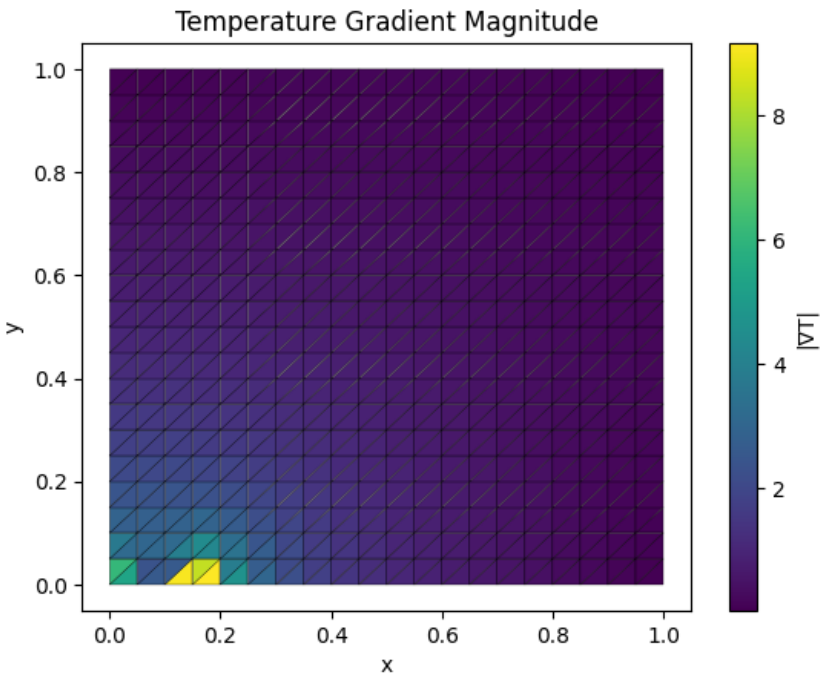


Figure 2: Gradient magnitude $|\nabla T|$ for constant heat source.

The gradient magnitude shows a relatively uniform distribution across the domain. This indicates that heat flows smoothly and steadily without abrupt changes in direction or intensity.

The absence of strong gradients reflects the uniform nature of the heat source, where temperature changes gradually over space.

### 8.1.3 Energy Density

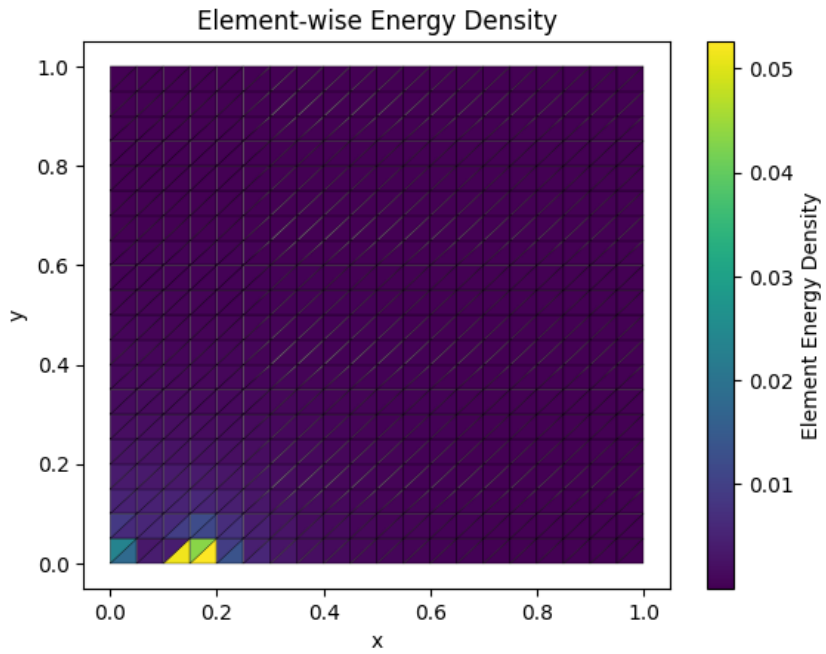


Figure 3: Energy density for constant heat source.

The energy density distribution closely follows the gradient magnitude. This is expected from the relation $E \propto |\nabla T|^2$, which implies that energy is directly determined by the temperature gradient.

As a result, the spatial patterns of energy density and gradient magnitude appear nearly identical. The difference lies only in scaling, with the energy distribution appearing slightly more pronounced due to the quadratic dependence.

### 8.1.4 Mesh Sensitivity Study

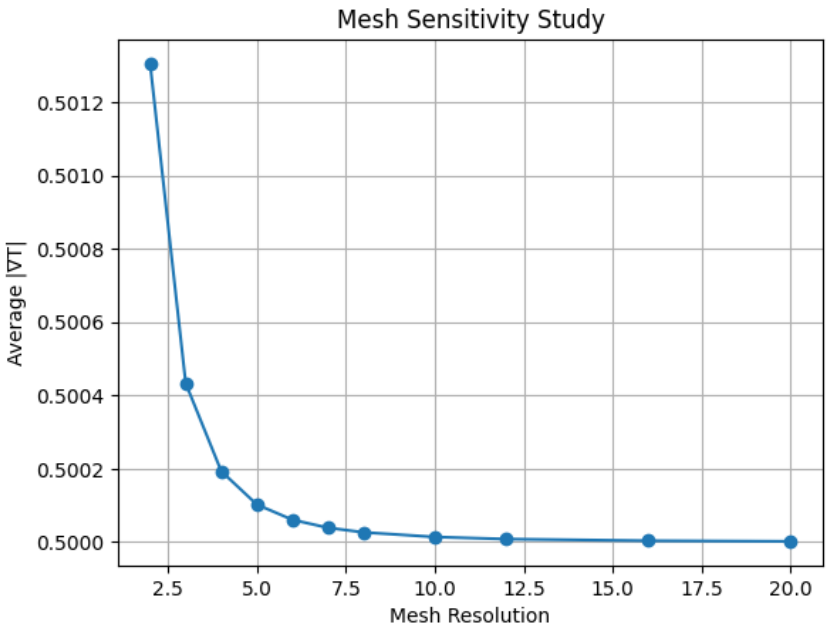


Figure 4: Mesh sensitivity study for constant heat source.

The mesh sensitivity study shows how the numerical solution changes with mesh refinement. For coarse meshes, the computed values differ noticeably due to insufficient resolution. As the mesh becomes finer, the solution stabilizes and approaches a constant value.

This behavior indicates convergence of the finite element solution and confirms that the numerical results are not dependent on the discretization beyond a certain resolution.

## 8.2 Gaussian Heat Source

### 8.2.1 Temperature Distribution

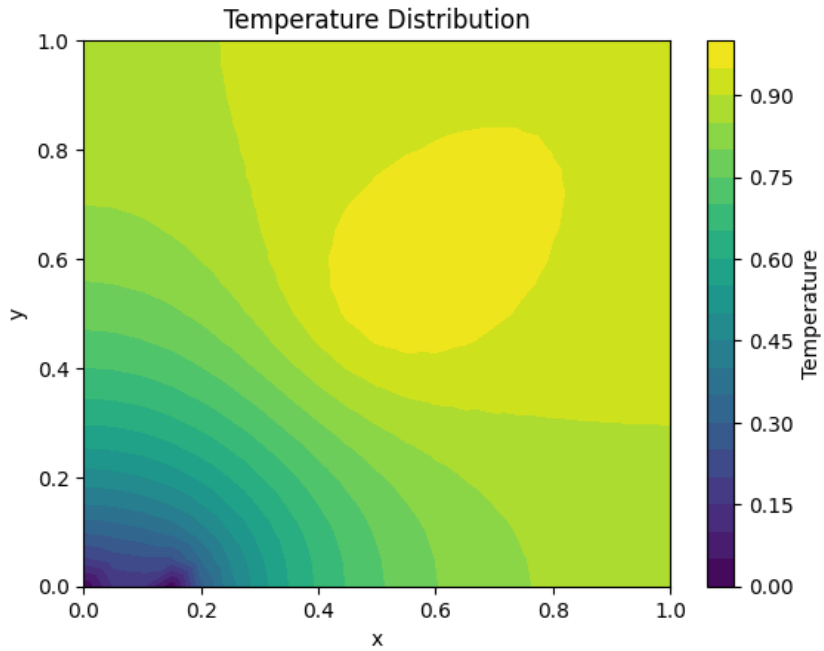


Figure 5: Temperature distribution for Gaussian heat source.

The temperature distribution for the Gaussian heat source exhibits a strong localized peak near the center of the domain. This reflects the concentration of heat generation in a small region.

Away from the center, the temperature decreases rapidly, indicating that heat diffuses outward from the localized source. This behavior closely resembles realistic physical scenarios involving concentrated heat input.

### 8.2.2 Gradient Magnitude

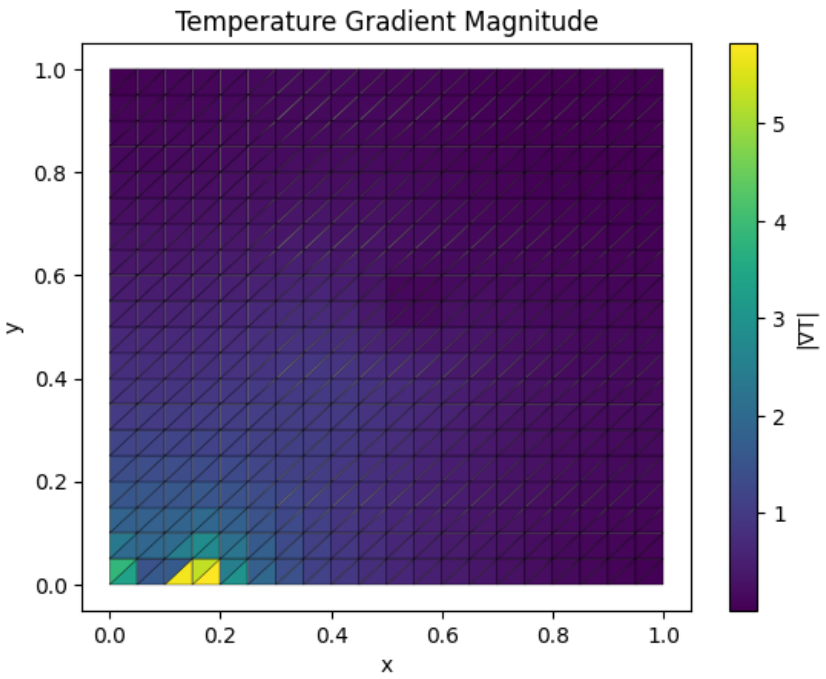


Figure 6: Gradient magnitude $|\nabla T|$ for Gaussian heat source.

The gradient magnitude is significantly higher near the center of the domain, where the temperature changes most rapidly. This indicates strong heat flow away from the hotspot.

As the distance from the center increases, the gradient decreases, reflecting the smoothing effect of diffusion.

### 8.2.3 Energy Density

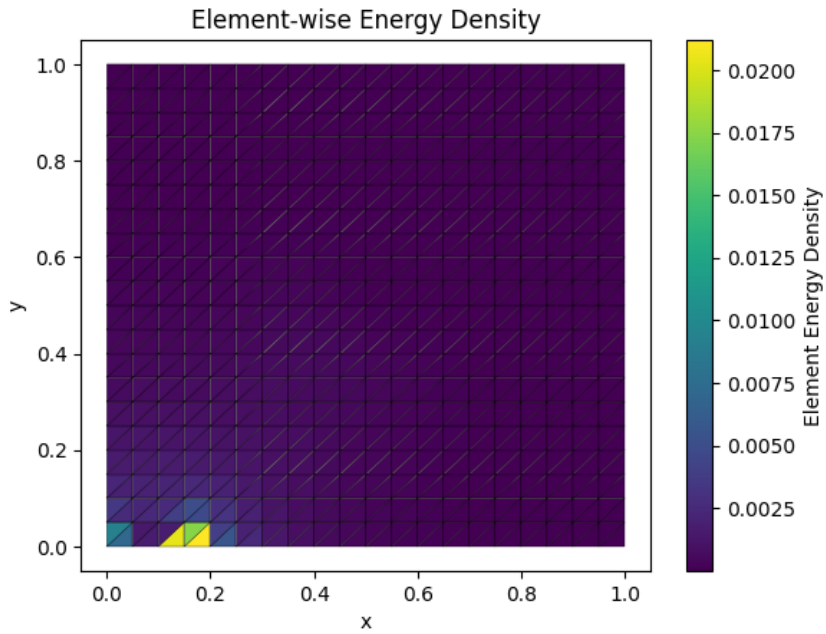


Figure 7: Energy density for Gaussian heat source.

The energy density is highly concentrated in the central region, corresponding to areas of large temperature gradients. As in the constant case, the energy distribution closely follows the gradient distribution.

The similarity between the gradient and energy plots arises from the quadratic dependence of energy on the gradient, resulting in nearly identical spatial patterns.

### 8.2.4 Mesh Sensitivity Study

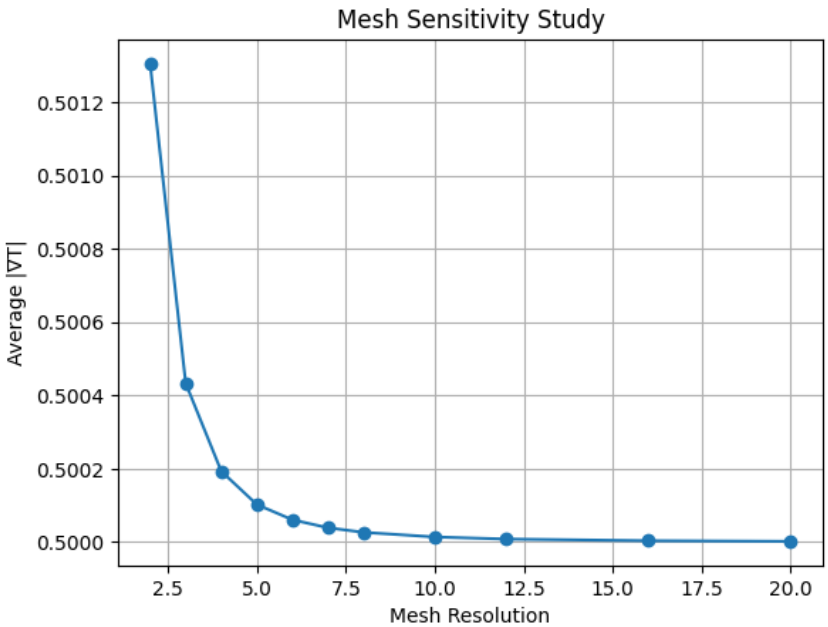


Figure 8: Mesh sensitivity study for Gaussian heat source.

The mesh sensitivity study for the Gaussian source also shows convergence as the mesh is refined. However, compared to the constant source, convergence is slightly slower due to the presence of sharp localized variations.

Finer meshes are required to accurately capture the behavior near the center, demonstrating the importance of mesh resolution in problems with strong spatial gradients.

# 9 Discussion

The results obtained in this study highlight the effectiveness of the finite element method in modeling steady-state heat conduction problems under different physical conditions.

The comparison between constant and Gaussian heat sources clearly demonstrates how the spatial distribution of the source term influences the solution. While the constant source leads to a smooth and globally distributed temperature field, the Gaussian source produces strong localization effects.

The analysis of gradient and energy density further enhances the physical understanding of the system. The close agreement between these quantities confirms the consistency of the numerical formulation and reflects their inherent mathematical relationship.

The mesh sensitivity study provides important validation by demonstrating convergence of the solution. This confirms that the numerical results are stable and physically meaningful.

Overall, the study shows that FEM is a powerful and flexible method for solving partial differential equations in complex physical systems.

# 10 Conclusion

In this project, the finite element method was applied to solve the two-dimensional steady-state heat equation with both constant and spatially varying heat sources.

A complete numerical framework was developed, including discretization of the domain, construction of element-level matrices, global assembly, and solution of the resulting system. Post-processing techniques were used to analyze temperature, gradient, and energy distributions.

The results demonstrate that the method accurately captures both uniform and localized heating effects. The Gaussian heat source provided additional insight into realistic scenarios involving concentrated energy input.

The mesh sensitivity study confirmed the convergence and reliability of the numerical solution, providing strong validation of the implementation.

This work establishes a solid foundation for further exploration of advanced topics in computational physics, including time-dependent problems, nonlinear systems, and multi-physics applications.